Job Dataset Exploratory Data Analysis (EDA)

In [13]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Introduction

This project performs Exploratory Data Analysis (EDA) on a job-related dataset. The dataset contains information about job titles, experience, education levels, skills count, and industries.

In [14]:
```
df=pd.read_csv("/content/job.csv")
```

Dataset Overview

The dataset contains multiple records related to different job roles and industries. It helps analyze employee experience, skills, education, and industry distribution.

What are the first five rows of the dataset?

In [15]:
```
df.head()
```

Out[15]:

	job_title	experience_years	education_level	skills_count	industry	com
0	AI Engineer	10	Bachelor	2	Healthcare	
1	Data Analyst	5	Bachelor	17	Telecom	
2	Frontend Developer	18	PhD	4	Media	
3	Business Analyst	19	PhD	13	Retail	
4	Product Manager	15	Bachelor	7	Manufacturing	

What are the column names in the dataset?

In [16]:
```
df.tail()
```

Out[16]:

	job_title	experience_years	education_level	skills_count	industry	cc
152983	DevOps Engineer	0	Diploma	1	Media	
152984	Data Scientist	17	Bachelor	15	Technology	
152985	Data Analyst	5	Diploma	5	Telecom	
152986	DevOps Engineer	0	Bachelor	7	Finance	
152987	Machine Learning Engineer	13	PhD	5	Governmen	

In [17]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 152988 entries, 0 to 152987
Data columns (total 10 columns):
 #   Column            Non-Null Count   Dtype
---  ------            --------------   -----
 0   job_title         152988 non-null  object
 1   experience_years  152988 non-null  int64
 2   education_level   152988 non-null  object
 3   skills_count      152988 non-null  int64
 4   industry          152988 non-null  object
 5   company_size      152987 non-null  object
 6   location          152987 non-null  object
 7   remote_work       152987 non-null  object
 8   certifications    152987 non-null  float64
 9   salary            152987 non-null  float64
dtypes: float64(2), int64(2), object(6)
memory usage: 11.7+ MB
```

What is the shape of the dataset?

In [18]:
```
df.shape
```

Out[18]: (152988, 10)

What is the basic information of the dataset?

In [19]:
```
df.describe()
```

Out[19]:

	experience_years	skills_count	certifications	salary
count	152988.000000	152988.000000	152987.000000	152987.000000
mean	9.993542	9.991496	2.491669	145646.673456
std	6.059693	5.474405	1.707387	37404.666237
min	0.000000	1.000000	0.000000	38779.000000
25%	5.000000	5.000000	1.000000	119285.000000
50%	10.000000	10.000000	2.000000	143346.000000
75%	15.000000	15.000000	4.000000	169435.000000
max	20.000000	19.000000	5.000000	325052.000000

Are there any missing values in the dataset?

In [20]:
```
df.isnull().sum()
```

Out[20]:

	0
job_title	0
experience_years	0
education_level	0
skills_count	0
industry	0
company_size	1
location	1
remote_work	1
certifications	1
salary	1

dtype: int64

In [28]:
```
df=df.dropna(axis=1)
```

In [29]:
```
df.isnull().sum()
```

Out[29]:

	0
job_title	0
experience_years	0
education_level	0
skills_count	0
industry	0

dtype: int64

Columns Check

What are the column names in the dataset?

In [39]:
```
print(df.columns)
```

```
Index(['job_title', 'experience_years', 'education_level', 'skills_count',
       'industry'],
      dtype='object')
```

What is the average experience of employees?

In [40]:
```
df["experience_years"].mean()
```

Out[40]:
```
np.float64(9.993541977148535)
```

What is the highest experience level?

In [41]:
```
df["experience_years"].max()
```

Out[41]:
```
20
```

What is the lowest experience level?

In [42]:
```
df["experience_years"].min()
```

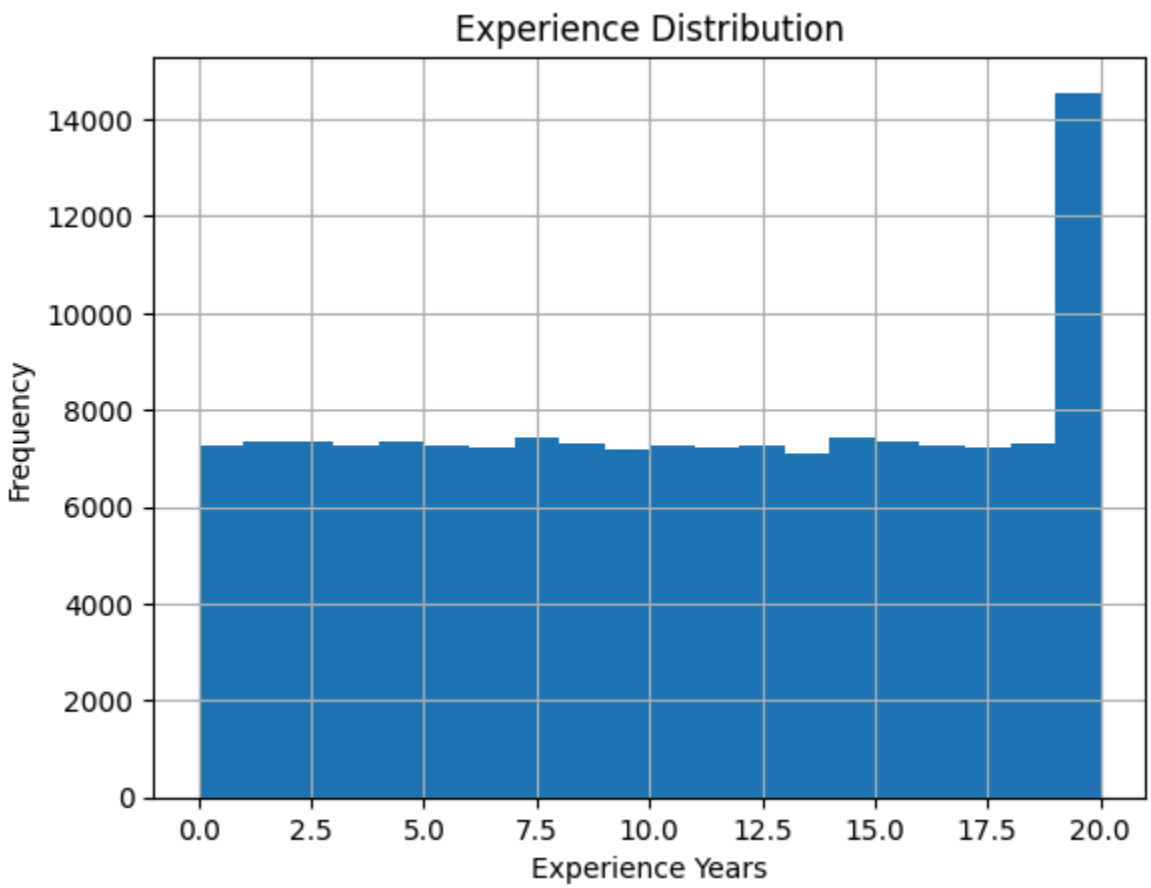
Out[42]:
```
0
```

Experience Histogram

Experience Distribution Histogram

In [43]:
```
import matplotlib.pyplot as plt

df["experience_years"].hist(bins=20)
plt.xlabel("Experience Years")
plt.ylabel("Frequency")
plt.title("Experience Distribution")
plt.show()
```



What is the average skills count?

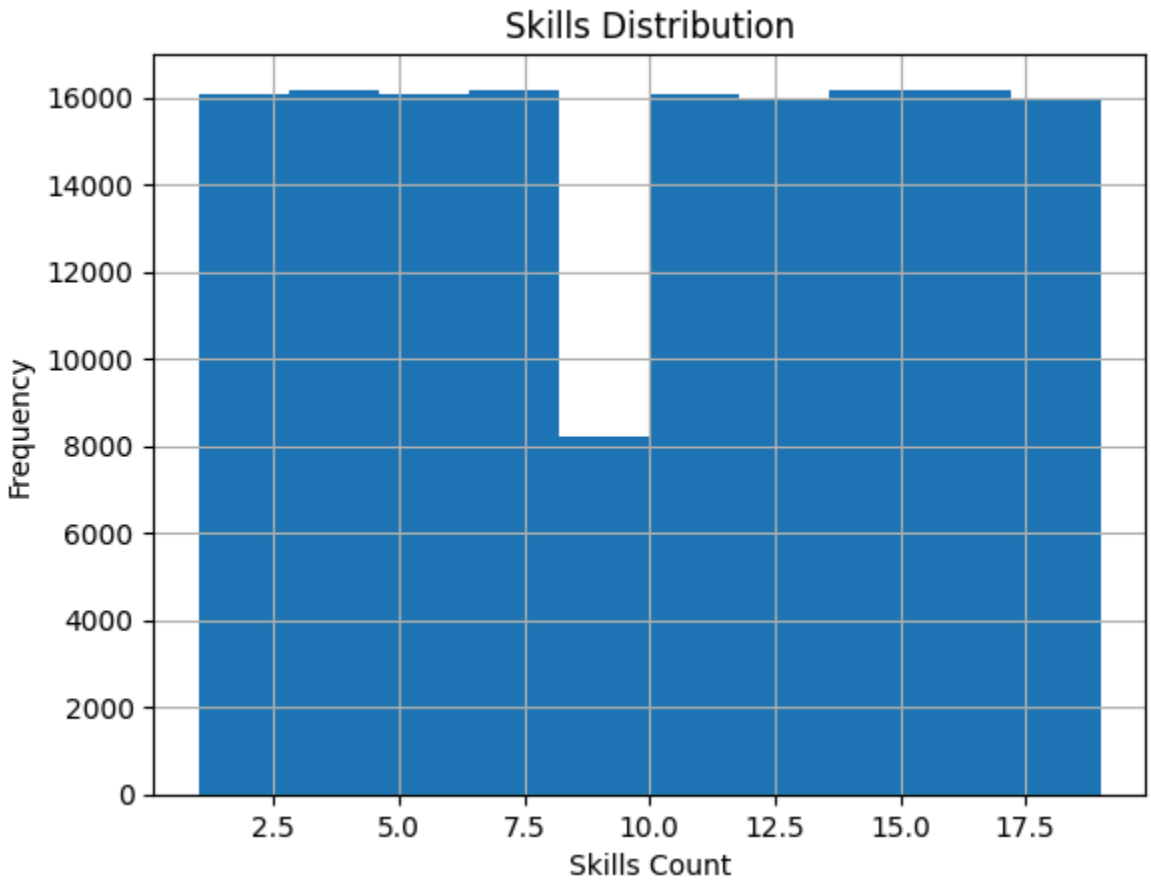
In [44]:
```
df["skills_count"].mean()
```

Out[44]:
```
np.float64(9.991496065050853)
```

Skills Count Distribution

In [45]:
```
df["skills_count"].hist(bins=10)
plt.xlabel("Skills Count")
plt.ylabel("Frequency")
```

```
plt.title("Skills Distribution")
plt.show()
```



Which job titles are most common?

In [46]:
```
df["job_title"].value_counts().head(10)
```

Out[46]:

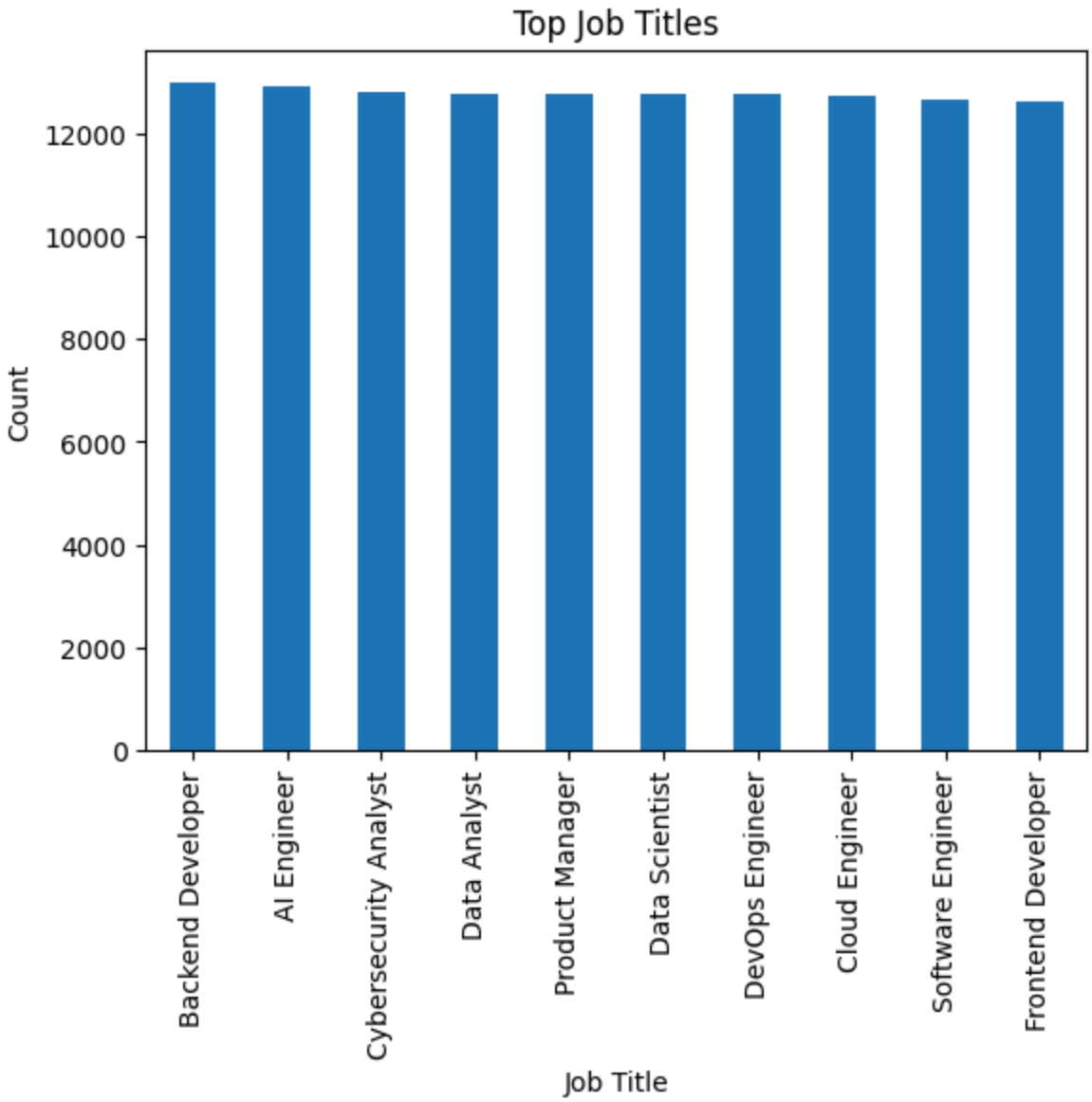
job_title	count
Backend Developer	12979
AI Engineer	12902
Cybersecurity Analyst	12790
Data Analyst	12779
Product Manager	12776
Data Scientist	12771
DevOps Engineer	12756
Cloud Engineer	12723
Software Engineer	12659
Frontend Developer	12631

dtype: int64

Top Job Titles Bar Chart

In [47]:

```
df["job_title"].value_counts().head(10).plot(kind="bar")
plt.xlabel("Job Title")
plt.ylabel("Count")
plt.title("Top Job Titles")
plt.show()
```



What is the distribution of education levels?

In [48]:
```
df["education_level"].value_counts()
```

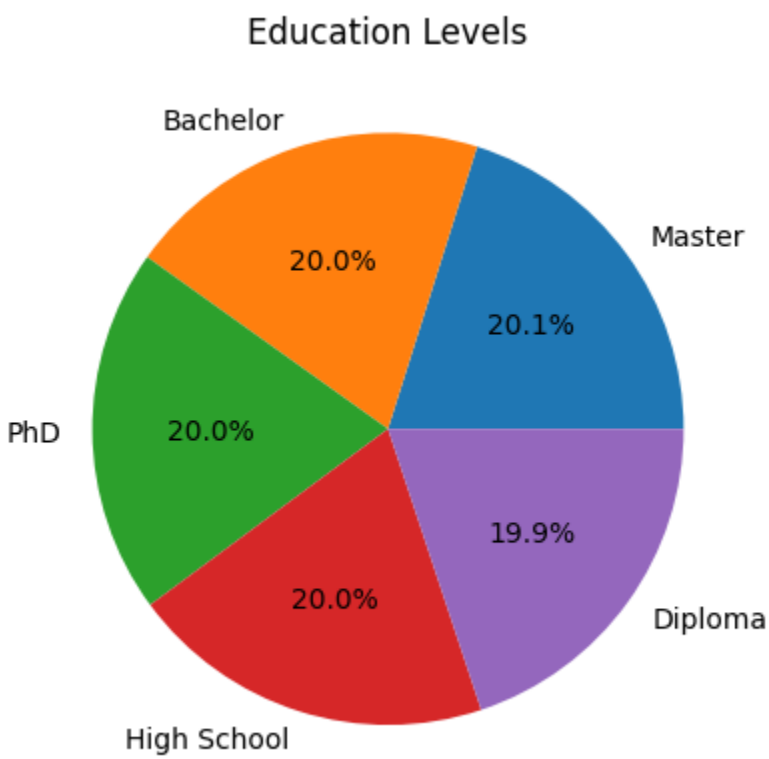
Out[48]:

education_level	count
Master	30787
Bachelor	30627
PhD	30580
High School	30559
Diploma	30435

dtype: int64

Education Level Pie Chart

In [49]:
```
df["education_level"].value_counts().plot(kind="pie", autopct="%1.1f%%")
plt.ylabel("")
plt.title("Education Levels")
plt.show()
```



Which industries have the highest number of employees?

In [50]:
```
df["industry"].value_counts()
```

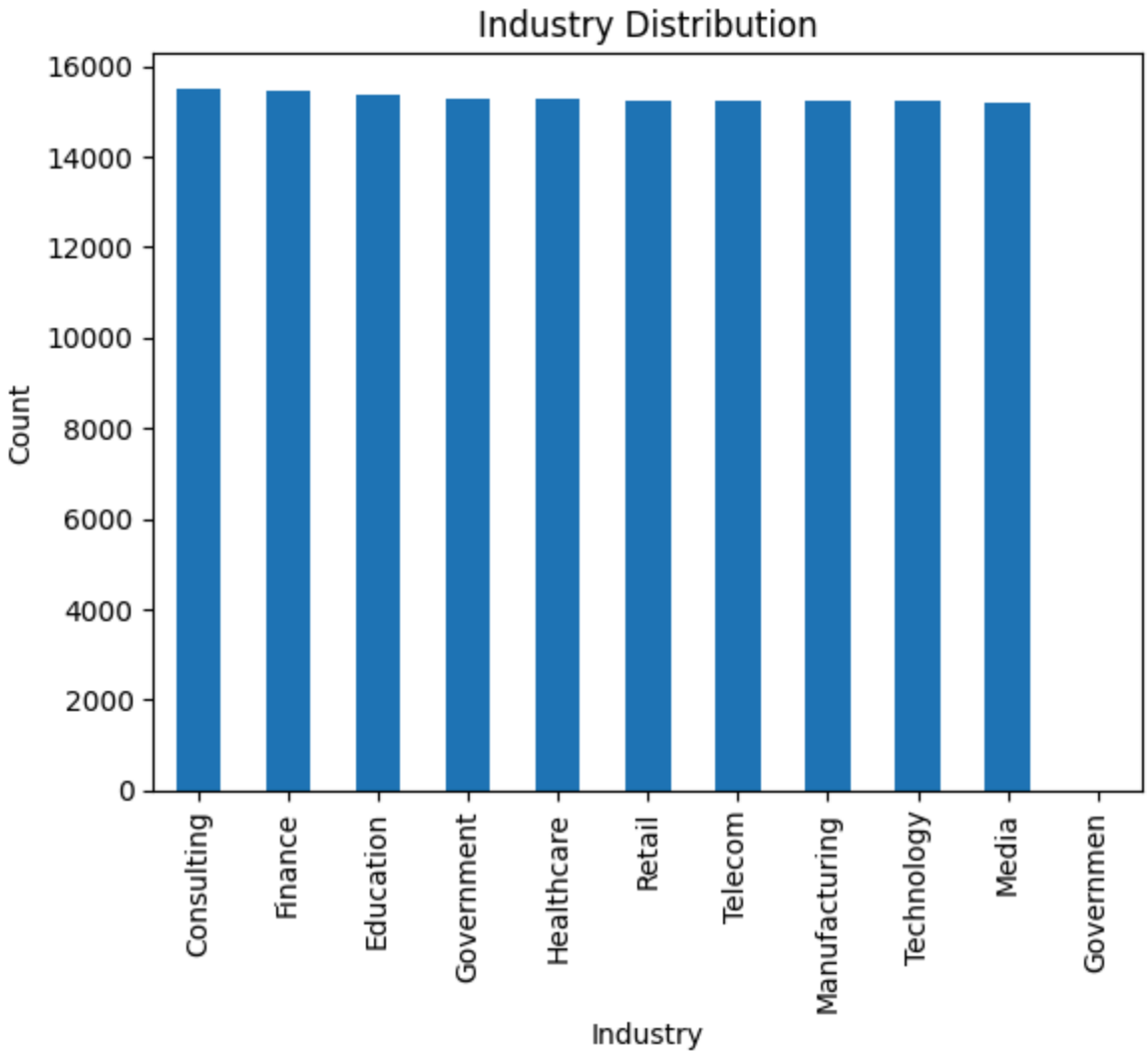
Out[50]:

industry	count
Consulting	15504
Finance	15468
Education	15381
Government	15297
Healthcare	15259
Retail	15236
Telecom	15219
Manufacturing	15219
Technology	15211
Media	15193
Governmen	1

dtype: int64

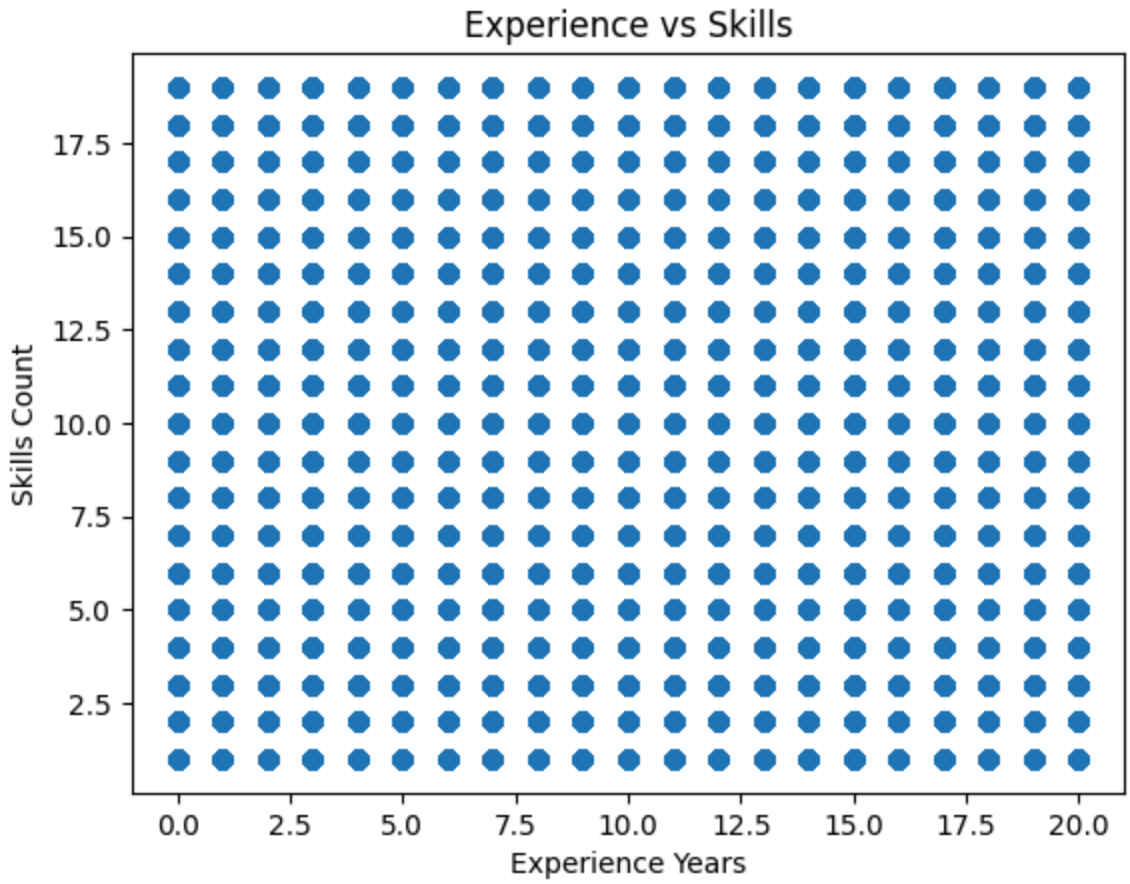
Industry Bar Chart

In [51]:
```
df["industry"].value_counts().plot(kind="bar")
plt.xlabel("Industry")
plt.ylabel("Count")
plt.title("Industry Distribution")
plt.show()
```



Experience vs Skills Scatter Plot

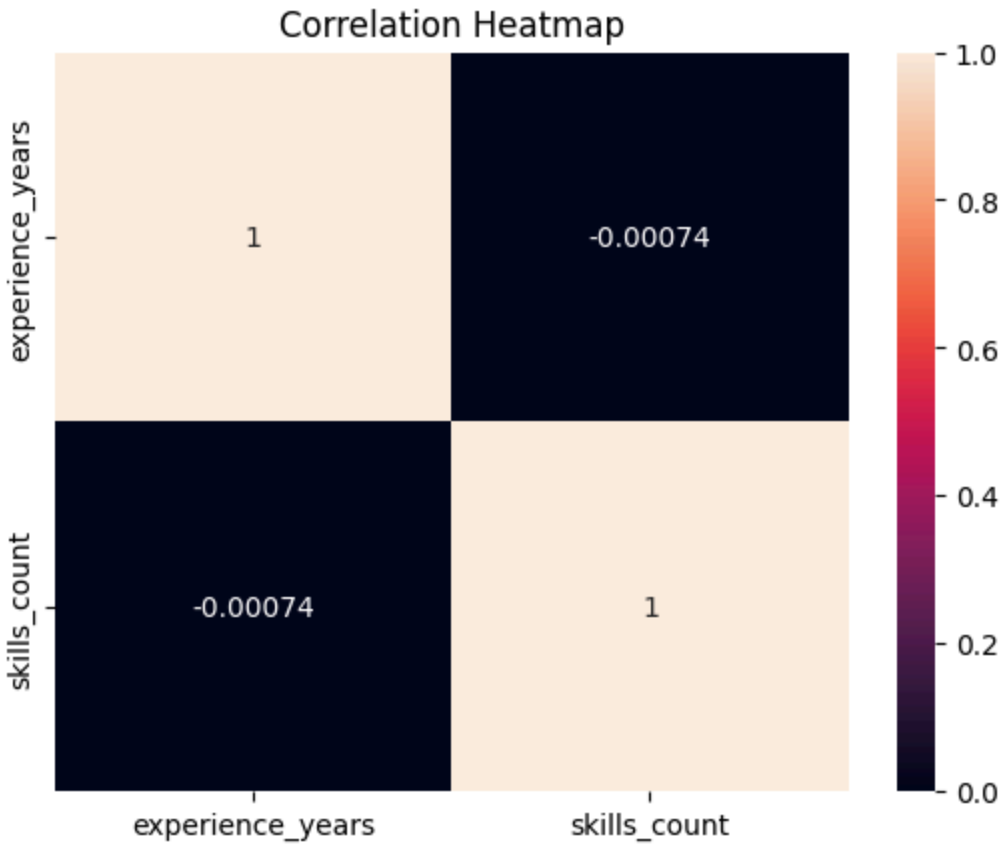
In [52]:
```
plt.scatter(df["experience_years"], df["skills_count"])
plt.xlabel("Experience Years")
plt.ylabel("Skills Count")
plt.title("Experience vs Skills")
plt.show()
```



Correlation Heatmap

In [53]:
```
import seaborn as sns

sns.heatmap(df.corr(numeric_only=True), annot=True)
plt.title("Correlation Heatmap")
plt.show()
```



Final Conclusion

The Exploratory Data Analysis (EDA) of the Job Salary Prediction Dataset provided useful insights into employee experience, skills, education, and industry trends. The analysis showed that experience, education level, and skills play an important role in career growth and salary opportunities. Data cleaning and visualization techniques helped in understanding patterns, relationships, and overall workforce distribution. This analysis can be useful for future salary prediction and machine learning models.

Graph Findings Experience Histogram: Most employees have medium-level experience. Skills Distribution: Employees with more skills tend to have better opportunities. Top Job Titles Chart: Some job roles are more in demand than others. Education Pie Chart: Higher education is common among employees. Industry Chart: Certain industries employ more workers. Scatter Plot: Experience and skills have a positive relationship. Heatmap: Some variables are positively correlated with each other.